



MP - 1 : Responding Refund Claims



Mesum Raza Hemani • 9:46 AM (Edited 9:59 AM)

40 points

Due May 17, 1:59 PM

Assignment: AI-Powered Refund Complaint System

Build a Streamlit application that uses a Large Language Model (LLM) to automate refund decisions for customer complaints.

Refund Policy

Refunds are allowed only if the customer submits the claim within of the delivery date.

Project Requirements

Your application should:

- Load a dataset of customer complaint tickets using Pandas and Display on Streamlit
- Process each ticket & Extract key information such as: Delivery Date, Claim Date [Use PyDantic with Langchain]
- Calculate "Number of Days Lapsed" [Important Hint: Use Python]
- Determine whether the refund should be APPROVED or REJECTED. [Important Hint: Use Python]
- **Bonus:** Based on Approval or Rejection Call LLM Once again to Generate a Relevant Email with Factual Information Extracted from Complaint such as Our Policy, Delivery Date, Claim Date, Number of Days Lapsed
- Store all outputs in a structured tabular format.
- Display processing progress and final results in Streamlit.

Grade Marks:

- Load Data from Excel in Pandas and Display on StreamLit: 2.5 Marks
- Building Langchain (PyDantic Parser) : 5 Marks
- Building Langchain (LCEL or RunnableSequence) : 5 Marks
- Correct Extraction of Delivery Date & Claim Date : 5 Marks
- Correct Calculation of Number of Days Lapsed & (Approve / Reject) : 5 Marks
- Bonus Marks - Generation of Relevant Response (Prompt Templates) : 2.5 Marks
- Bonus Marks - Generation of Relevant Response (Prompt Results) : 2.5 Marks
- Storing All the Data into the Pandas Structured DataFrame: 2.5 Marks
- Display Data on Streamlit: 2.5 Marks
- Clean Structured Code : 5 Marks

Submission:

- Submit the Code [.py file only (Dont submit your Keys or env files)]
- A Running Video Demo with your Audio to Explain App & Functionality



[Refund Claims_Data.x...](#)

Microsoft Excel

[Iterate Through Rows...](#)

YouTube video • 4 minutes



Your work

Assigned

[+ Add or create](#)

Mark as done

 Private comments[Add comment to Mesum Raza Hemani](#) Class comments [Add comment](#)